

SolarGuard AI

Economic Analysis & Business Model — Lean Edition

\$200K Seed · Off-the-Shelf Hardware · 30 Farms · NPV +\$4M · Profitable Year 1

1. Executive Summary

SolarGuard AI detects solar panel microcracks before they cause power loss, using commercial off-the-shelf hardware and a proprietary AI model. The lean architecture — commercial EL cameras, edge compute modules, and fine-tuned open-source ML — means the business can launch with a USD 200K seed, is profitable from Year 1, and reaches a +USD 4M NPV over 10 years.

1.1 Key Metrics

Metric	Value
Seed Investment	USD 200K (lean bootstrapped)
Hardware Approach	Off-the-shelf (EL camera + Jetson Nano) — 25% COGS
Team (Year 1)	2 people, fully remote
Annual SaaS Fee	USD 47,500/farm/yr
One-time HW + Impl	USD 400,000/farm
Max Farms (Year 10)	30 farms
Year 10 SaaS ARR	USD 1.43M/yr
10-Year Revenue	USD 18.0M
EBITDA Breakeven	Year 1 (EBITDA +\$203K)
NPV @ 15% WACC	USD +4.0M
Customer 25-yr ROI	~250% on SolarGuard cost

2. Why USD 200K Is Enough

The key insight is that custom hardware is unnecessary. Commercial electroluminescence (EL) cameras and NVIDIA Jetson edge compute modules are available off the shelf for USD 6K–16K per site. Combined with a fine-tuned open-source ML model (PyTorch / TensorFlow), the technical stack costs a fraction of a traditional hardware startup.

2.1 Year 1 Budget Breakdown

Category	Amount	%	Notes
AI/ML Fine-tuning	\$100K	50%	Remote AI engineer + cloud GPU (AWS/GCP) for 12 months
Pilot Hardware Kit	\$30K	15%	EL camera (\$10K) + Jetson Nano (\$1K) + install for 1 farm
Operations & Sales	\$40K	20%	Founder minimal salary + travel + BD
Legal / IP	\$20K	10%	Company incorporation, NDA templates, IP filing
Contingency	\$10K	5%	Buffer
Total	\$200K	100%	

Year 1 revenue from the first farm: USD 447.5K (USD 400K hardware + USD 47.5K SaaS). Year 1 EBITDA: +USD 203K. The seed is fully recovered within the first client contract.

3. Scientific Foundation

Microcrack-induced power loss follows a logistic S-curve: $L(t) = K / (1 + A \cdot \exp(-r \cdot t))$, where $K=0.20$, $A=24$, and r is the propagation rate constant. For Al Kharsaah (800 MWp, Qatar, generation 1,971,542 MWh/yr at QAR 0.0571/kWh), the base case ($r=0.254$) yields a 25-year loss of QAR 21.6M = USD 5.93M per farm.

Scenario	Rate r	25-yr USD Loss	70% Recoverable
Conservative	0.200	USD 3.00M	USD 2.10M
Base Case	0.254	USD 5.93M	USD 4.15M
Aggressive	0.350	USD 6.35M	USD 4.45M

4. Platform Architecture

- EL Camera (off-the-shelf): Commercial units (e.g., Solibot, Greateyes) capture electroluminescence images of panels. Cost: \$5K–15K per farm.
- Edge Compute (Jetson Nano): On-site inference — no cloud latency, no data privacy issues. Cost: ~\$500–1K per site.
- AI Model: Fine-tuned ResNet/EfficientNet on public microcrack datasets. Detects crack patterns invisible to human inspection.
- SaaS Dashboard: Web + mobile interface showing per-panel health, predicted degradation, and maintenance queue.
- ROI Reporting: Monthly automated PDF showing MWh recovered vs. subscription cost.

5. Business Model & Financials

5.1 Revenue Streams

Stream	Price	COGS	Gross Margin
Annual SaaS	USD 47,500/farm/yr	30%	70%
Hardware + Implementation	USD 400,000/farm (one-time)	25% (off-the-shelf)	75%

Pricing rationale: the customer's 25-year SaaS cost (USD 1.19M) is only 20% of their base-case loss (USD 5.93M). They keep 80% of recoverable savings. Customer gains approx USD 3M net over 25 years on a USD 1.19M investment (250% ROI).

5.2 Farm Rollout

Year	Cumulative Farms	New Farms	Phase
1	1	1	Qatar anchor pilot (Al Kharsaah)
2	2	1	Second pilot site
3	3	1	Third site; model fully validated
5	8	3	MENA expansion; channel partners
7	17	5	Scale across MENA
10	30	4	Full SOM — USD 1.43M SaaS ARR

5.3 10-Year Financial Projections

Year	Farms	SaaS ARR	Total Revenue	Total Costs	EBITDA	Margin
Year 1	1	\$0.048M	\$0.448M	\$0.244M	\$0.203M	45.4%
Year 2	2	\$0.095M	\$0.495M	\$0.298M	\$0.197M	39.7%
Year 3	3	\$0.142M	\$0.542M	\$0.323M	\$0.220M	40.5%
Year 5	8	\$0.380M	\$1.580M	\$0.614M	\$0.966M	61.1%
Year 7	17	\$0.807M	\$2.808M	\$0.992M	\$1.815M	64.7%
Year 10	30	\$1.425M	\$3.025M	\$1.078M	\$1.948M	64.4%

Key assumptions: SaaS COGS 30%; hardware COGS 25% (off-the-shelf); R&D declines from \$100K/yr (Yr 2) to \$50K/yr (Yr 4+); Ops grows from \$50K to \$200K/yr. Profitable every year from Year 1.

6. Investment Returns

Metric	Value
Seed Required	USD 200K
WACC	15% (frontier market tech)
NPV @ 15% (10-year)	USD +4.01M
Payback Period	Year 1 (within first client contract)
10-yr Cumulative EBITDA	USD 11.1M
Year 10 SaaS ARR	USD 1.43M (fully recurring)

With a lean cost structure, the company is self-funding from Year 1. A Series A at Year 5 can accelerate beyond 30 farms without being required for survival — it is purely growth capital.

7. Use of USD 200K Seed

Category	Amount	%	Purpose
AI / ML Engineering	\$100K	50%	1 remote AI engineer, cloud GPUs, model development
Pilot Hardware	\$30K	15%	EL camera + Jetson + installation for first farm
Operations & BD	\$40K	20%	Founder cost, travel, sales outreach
Legal / IP	\$20K	10%	Incorporation, IP filing, contracts
Contingency	\$10K	5%	Buffer

8. Risk Factors

Risk	Likelihood	Mitigation
HW vendor dependency	Low	Multiple commercial EL camera suppliers; Jetson is widely available
AI model accuracy	Medium	Validated on public microcrack datasets; pilot refines on real data
Slow adoption	Medium	ROI-first pricing; customer gains \$3M net over 25yr on \$1.19M spend
Competitive entry	Low–Medium	Proprietary dataset moat from Al Kharsaah; first-mover in MENA
Tariff change	Low	MENA diversification from Year 3; fixed SaaS contracts

9. References

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